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Weight stigma predicts reduced psychological wellbeing and weight gain among sexual minority men: A 12-month longitudinal cohort study using random intercept cross-lagged panel models

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ABSTRACT

Weight stigma among sexual minority men predicts negative health outcomes; however, existing research is cross-sectional. Therefore, we longitudinally examined the presence, strength, and direction of relationships between weight stigma and health outcomes in a multi-national cohort of 2953 sexual minority men (M_{age} = 37.70, SD_{age} = 13.70). Participants reported their Body Mass Index (BMI), experiences of weight discrimination, internalized weight bias, and psychological quality of life at three time-points over 12 months (~August 2019–20). Analyses revealed feedback loops of internalized weight bias with BMI and psychological quality of life: over time, increases in internalized weight bias were associated with small BMI increases (β = 0.07) and small decreases in psychological quality of life (β = 0.12). Reciprocally, increases in BMI and decreases in psychological quality of life were associated with increases in internalized weight bias (βs = 0.18). Experiences of weight discrimination were not longitudinally predictive and were associated with higher BMI (β = 0.29), greater internalized weight bias (β = 0.29), and lower psychological quality of life (β = 0.26) between-persons/cross-sectionally only. Weight stigma is longitudinally and reciprocally associated with reduced psychological wellbeing and weight gain among sexual minority men. Internalized weight bias is a particularly promising intervention target.

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1. Introduction

Weight stigma, the devaluation of a person due to their body weight, disproportionately affects people with higher body weights (Griffiths et al., 2018a; Meadows & Calogero, 2018). Experiencing weight discrimination, and the self-devaluation that results from internalizing negative weight stereotypes (i.e., *internalized weight bias*), are associated with negative mental health outcomes, including depression, anxiety, and body dissatisfaction (Emmer et al., 2020; Pearl & Puhl, 2018). Recent studies with sexual minority men (i.e., men who identify as non-heterosexual identities, including gay and bisexual) indicate that this population may be more vulnerable to weight stigma phenomena than heterosexual men (Austen et al., 2020; Puhl, Himmelstein, Pearl, Wojtanowski & Foster, 2019), emphasizing the need for further research among this population. In this paper, we present a longitudinal examination of how two weight stigma constructs, experiences of weight discrimination and internalized weight bias, relate with BMI and psychological wellbeing among sexual minority men.

1.1. Weight stigma: experiences and internalization

Experiences of weight discrimination and internalized weight bias are related but distinct constructs. Recent review papers indicate that both are more robustly associated with negative mental health outcomes than physical health outcomes (Emmer et al., 2020; Pearl & Puhl, 2018). Understanding how experiences of weight discrimination and internalized weight bias relate to one another, and with other variables like BMI and psychological wellbeing, is necessary for intervention development.

Various cross-sectional models have explored internalized weight bias as a mechanism underlying the relationship of experiences of weight discrimination with poorer psychological wellbeing (Austen et al., 2020; Hayward et al., 2018). Stated differently, experiences of weight discrimination are hypothesized to increase one's level of internalized weight bias, which subsequently leads to diminished psychological wellbeing. Some studies have investigated BMI as another independent variable in these relationships, such

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that higher BMIs are associated with poorer psychological wellbeing through more frequent experiences of weight stigma (e.g., Griffiths et al., 2018a; Zuba & Warschburger, 2017). However, growing evidence suggests that weight stigma experiences and internalization are associated with adverse outcomes even after controlling for BMI (Pearl & Puhl, 2018; Emmer et al., 2020), suggesting that weight stigma is associated with outcomes like poorer psychological wellbeing across the weight spectrum.

While various studies have examined the ways weight stigma experiences and internalization operate together, with other variables like BMI, to facilitate poorer psychological wellbeing, some experimental evidence indicates that internalized weight bias and experiences of weight discrimination also independently predict negative outcomes (Pearl & Puhl, 2016). Thus, to highlight precise intervention targets, studies can focus on how these constructs relate to one another, but also how they independently relate to outcomes.

1.2. Sexual minority men are more vulnerable to weight stigma

Sexual minority men are underrepresented in weight stigma research, despite being particularly vulnerable to weight stigma. For example, Austen and colleagues (2020) found that bisexual men reported more frequent experiences of weight stigma than both gay and heterosexual men, which the authors attributed (in part) to the increased societal stigma that bisexual men face from heterosexual and sexual minority groups. Two recent studies demonstrated that, compared to heterosexual men, sexual minority men report higher levels of internalized weight bias (Austen et al., 2020; Puhl et al., 2019). Similarly, Foster-Gimbel and Engeln (2016) found that gay men reported higher levels of antifat bias (i.e., endorsing negative stereotypes about fat people) than heterosexual men, both in relation to themselves and when evaluating potential romantic partners.

The increased vulnerability of sexual minority men to weight stigma may be attributable to the potent appearance pressures they face. For example, compared to heterosexual men, sexual minority men idealize body types that are more visibly muscular; that is, bodies that are both low in body fat and high in muscularity (Morrison et al., 2004; Strübel & Petrie, 2019). These narrow body ideals saturate magazines, dating applications, and social media targeted at sexual minority men, emphasizing the ubiquity of these pressures in everyday life (Anderson, Holland, Koc & Haslam, 2018; Jankowski, Fawkner, Slater & Tiggemann, 2014; Whitesel, 2014). These appearance pressures may compound the stigma associated with having a sexual minority identity – one factor that underlies the increased rates of mental ill health among sexual minorities (Plöderl & Tremblay, 2015). Overall, examinations of weight stigma among sexual minority men are warranted, given their vulnerability to weight stigma and mental ill health.

1.3. Longitudinal examinations can help identify intervention targets

Existing weight stigma studies are predominantly cross-sectional, and studies of weight stigma among sexual minority men are exclusively cross-sectional. Separating the between- (trait-like) and within-person (state-like) variation in the relationships of weight stigma and its outcomes via longitudinal assessments is necessary for understanding these processes and ameliorating negative outcomes. When examining relationships among psychological phenomena, failure to partition within- and between-person variance can facilitate inaccurate inferences regarding the strength, presence, and direction of causal relationships, and mask precise intervention targets (Hamaker, Kuiper, & Grasman, 2015).

As previously mentioned, cross-sectional research highlights associations among weight stigma and outcomes, finding, for example, that higher average levels of internalized weight bias are associated with higher BMIs, and poorer psychological quality of life (Pearl & Puhl, 2018). This between-person data can be harnessed to determine what specific groups within a population may be vulnerable to experiencing and internalizing weight stigma (e.g., people with higher BMIs). However, to develop effective interventions, it is also crucial to examine how weight stigma may operate over time at an *individual level* (i.e., within-persons). For instance, if someone experiences more weight discrimination *than is typical for them* (i.e., their own average), do they then internalize more weight bias, or gain more weight, than is typical for them at a later timepoint? Understanding longitudinal within-person relationships is particularly relevant to weight stigma, where interventions are often implemented at an individual level to help people cope with their experiences and reduce negative outcomes (e.g., Acceptance and Commitment Therapy [Griffiths et al., 2018b]; Cognitive Behavioural Therapy [Pearl et al., 2020]).

Using longitudinal methods can also help clarify the direction in which weight stigma and its outcomes operate. Existing studies predominantly examine weight stigma as a unidirectional predictor of outcomes; however, several authors have proposed that these relationships may be bidirectional (e.g., Hayward et al., 2018; Major et al., 2018; Pearl & Puhl, 2018). For instance, does internalized weight bias facilitate decreased psychological wellbeing, or do reductions in psychological wellbeing make people more vulnerable to internalizing weight bias – or both? Further, while it is apparent that weight gain (i.e., a BMI increase) may lead to further internalization of weight bias or experiences of weight discrimination (Griffiths et al., 2018a; Zuba & Warschburger, 2017), the opposite may also be true. For instance, Major and colleagues (2018) propose that being more aware of negative weight stereotypes may make individuals avoid 'health-related' behaviours (e.g., exercise, healthy eating) out of fear of stigmatization, thereby facilitating weight gain. Supporting this, Vartanian and colleagues (2018) found that people higher in internalized weight bias, those with higher previous experiences of weight stigma, and female participants were less likely to express motivation to exercise and eat healthily following a weight stigma experience (as reported via ecological momentary assessment surveys).

1.4. The current study

This paper presents five random intercept cross-lagged panel models (RI-CLPMs; Hamaker et al., 2015) examining the relationships of experiences of weight discrimination and internalized weight bias with psychological quality of life and BMI among a large cohort of sexual minority men (N = 2953). Fig. 1 presents our first model, which examines the relationship of internalized weight bias with BMI over three timepoints. Models two through five examine other pairings among our study variables. Although we examine bidirectional relationships, we have proposed within-person hypotheses only for the direction of relationships that is assessed by the majority of existing literature – that is, weight stigma constructs predicting outcomes.

We hypothesized that: (1) higher average internalized weight bias would be associated with higher BMIs, and poorer psychological quality of life, (2) within-person increases in internalized weight bias would be associated with higher BMIs and poorer psychological quality of life within-persons at the next timepoint, (3) more experiences of weight discrimination than average would be associated with greater average internalized weight bias, higher BMIs and poorer psychological quality of life, and (4) more frequent experiences of weight discrimination within-persons would be associated with higher levels of internalized weight bias, increases in BMI, and decreases in psychological quality of life within-persons at the subsequent timepoint.

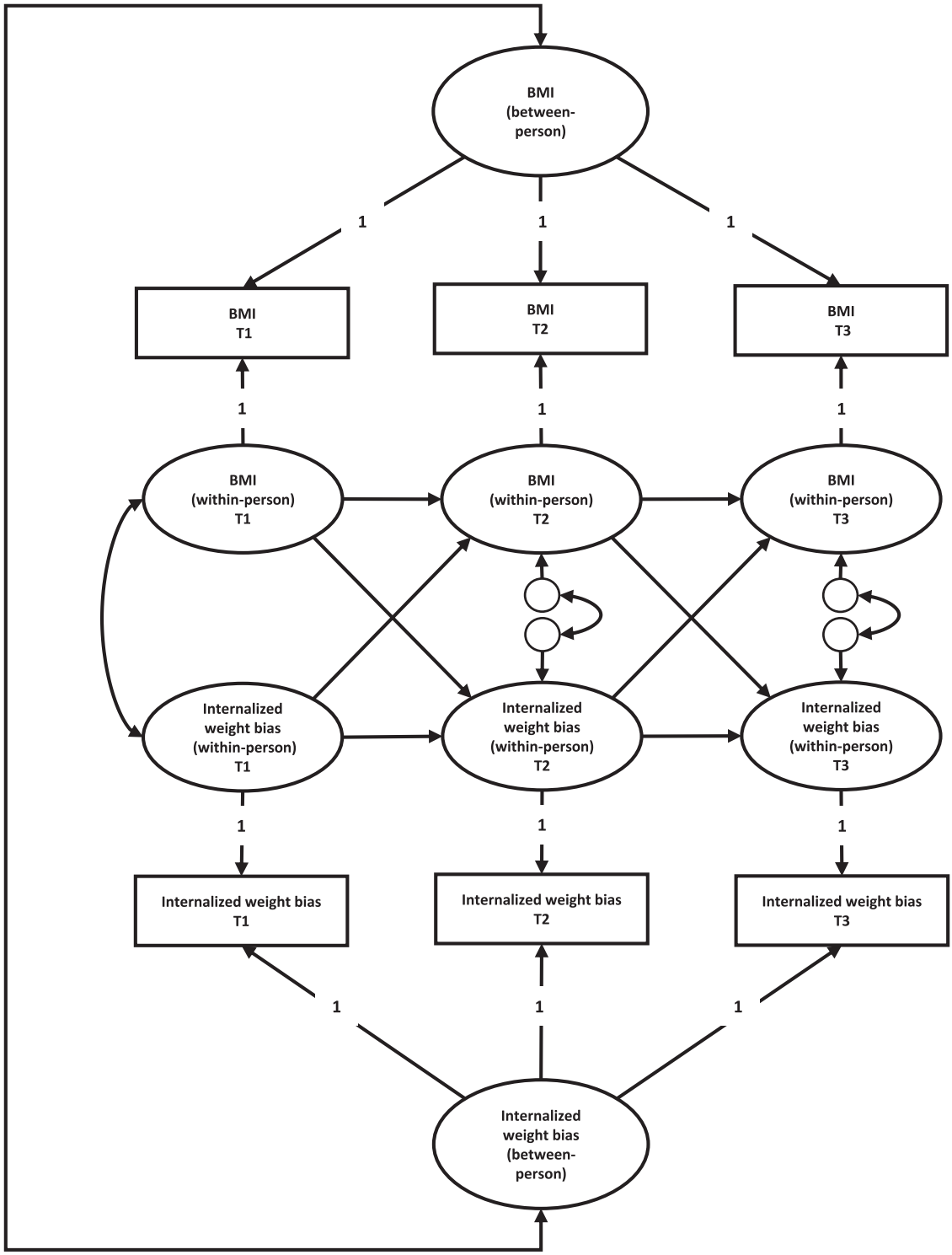

Fig. 1. Our random-intercept cross-lagged panel model (RI-CLPM) assessing the relationships of Body Mass Index (BMI) with internalized weight bias over three timepoints. Observed scores (i.e., participant scores on internalized weight bias and BMI at T1, T2, and T3) are partitioned into within-person (i.e., state) and between-person (i.e., trait) variation. The cross-lagged paths (e.g., internalized weight bias at T1 predicting BMI at T2) and auto-regressive paths (e.g., internalized weight bias at T1 predicting internalized weight bias at T2) represent the longitudinal relationships between the within-person components (e.g., someone's deviation from their own average BMI at a given timepoint). The random intercepts control for the between-person variation in observed scores (e.g., average BMI). In our models including SES, race, and age as covariates, random intercepts were regressed onto the covariates. T1, T2, and T3 = time-points 1, 2, and 3, respectively. The lag between each time-point is approximately 6 months (see Supplementary Fig. 2).

2. Method

2.1. Open science commitment

Our deidentified study data (raw and clean), materials, and code are freely available on the Open Science Framework (https://osf.io/tb25g/). We warmly encourage researchers to avail themselves of these resources.

2.2. Cohort and procedure

Participants were sexual minority men recruited to Gay Bodies Worldwide: a cohort study of body image phenomena among sexual minority men that launched in 2019 and concludes in 2024. Participants were recruited via advertisements disseminated on Grindr – a smartphone dating/hook-up application for sexual minority men, transgender, and queer individuals. Advertisements appeared as either static or animated full-screen images on users' Grindr interface. Messages that accompanied advertisement images included "What are you attracted to?", and "Do you think about how you look?", along with an invitation to participate in a 10 minute survey as part of Gay Bodies Worldwide. The full range of advertisements are viewable on our Open Science Framework page. Advertisements appeared nationwide to users in Australia, the United States of America, the United Kingdom, and Canada.

Follow-up data collection occurred at six-month intervals via email and SMS (phone text-message) invites. In the study presented here, the three timepoints correspond to the following time periods: Time 1 = August–November 2019, Time 2 = February–May 2020, and Time 3 = August–November 2020. The data presented here comes from the weight stigma sub-study of Gay Bodies Worldwide. In total, 2973 participants in Gay Bodies Worldwide were randomized to the weight stigma sub-study. From these 2973 participants, we excluded data from individuals younger than 18 years old due to ethics requirements (n = 20). Furthermore, at each timepoint, we excluded participants who accessed the survey but did not proceed past the consent page ([we requested consent from participants at each timepoint]: n's: Time 1 = 83, Time 2 = 59, Time 3 = 41) together with participants who identified as heterosexual (n's: Time 1 = 2, Time 2 = 3. Time 3 = 5). Therefore, the post-exclusions sample size was 2953. The sample size at each time-point was 2259 at Time 1, 1935 at Time 2, and 1731 at Time 3. The reason why the final sample size (2953) differs from the Time 1 sample size (2259) is because some participants did not provide data at Time 1 but did provide data at Time 2 and 3. For a clarifying visual representation of the above figures, please see Supplementary Fig. 1. Characteristics of participants who dropped out of at least one subsequent timepoint are provided in Supplementary Table 1. In total, 1072 (36%) provided data for all three timepoints. We modelled participants' missing data such that every participant who provided at least one timepoint of data was represented in our longitudinal analyses – a critical necessity to avoid biased results in longitudinal research (see Data Analyses for more information). Participants were volunteers and unremunerated. The study procedure was approved by the University of Melbourne Human Research Ethics Committee (ID: 1853237.4)

2.3. Measures

2.3.1. Outcomes

Experiences of weight discrimination were measured with an adapted version of the Everyday Discrimination Scale (EDS; Williams, Yu, Jackson & Anderson, 1997). Consistent with previous research (Hunger, Dodd, & Smith, 2020), we modified the survey descriptor to reflect weight-related discrimination: "In your day-to-day life, how often do any of the following things happen to you *because of your weight*?". Participants responded to items (e.g., "You are treated with less respect than other people are") using a 5-point Likert scale (1 = never, 5 = all the time). This version of the EDS demonstrated excellent internal reliability in our study (Cronbach's αs at every timepoint = 0.92, 95% confidence interval [CI; 0.92, 0.93]). We note that we also calculated McDonald's omegas (ω) for all study measures; however, the ω point estimates and 95% CIs were nearly identical to our αs (< 0.01 discrepancy in all cases). Thus, for brevity, we only report our α estimates.

Internalized weight bias was assessed with the 11-item Weight Bias Internalization Scale – Modified (WBIS-M; Pearl & Puhl, 2014). Participants responded to items (e.g., "I hate myself for my weight") using a 7-point Likert-type scale (1 = strongly disagree, 7 = strongly agree). The WBIS-M demonstrated excellent reliability in this study (α's at every timepoint = 0.94, 95% CI [0.93, 0.94]).

Psychological quality of life was assessed with the 24-item 'Mental Super Dimension' (MSD) subscale of the Assessment of Quality of Life – 8 Dimension measure (Richardson, Iezzi, Khan & Maxwell, 2014). The MSD measures various facets of psychological quality of life (e.g., mental health, relationships) in the past week. Participants responded on Likert-type scales (response options varied depending on the question). Items were totalled to provide a composite score (minimum = 24, maximum = 125), wherein higher scores denote decreased psychological quality of life. The MSD demonstrated strong internal reliability in our study (α's at every timepoint = 0.95, 95% CI [0.95, 0.96]).

Participants self-reported their height and weight using their preferred metric, which was later used to calculate body mass index (BMI, kg/m^2). Notably, self-report height and weight measures have demonstrated very good concordance with objective measures (r's =.98 and.99; Pursey, Burrows, Stanwell & Collins, 2014).

2.3.2. Demographic and covariate measures

Participants self-reported their age, race, and socioeconomic status (SES). To ensure a fulsome capture of sexual orientation, participants reported their sexual identity, attractions, and experiences (Mustanski, Birkett, Greene, Rosario, Bostwick & Everett, 2014). Sexual identity was operationalized by one item: "What best describes your sexual identity?". Response options included: asexual, bisexual, gay, heterosexual/straight, pansexual, and questioning, and a free-text box was available for other responses. Sexual experiences were assessed with one item: "Who have you had sexual experiences with?". Sexual attraction was assessed with one item: "Who are you sexually attracted to?". Response for sexual experiences and attraction included options ranging from: "only with/to males, never with/to females" to "only with/to females, never with/to a male", and a free-text box was available for other responses. SES was assessed with the one-item MacArthur Scale of Subjective Social Status (Adler, Epel, Castellazzo & Ickovics, 2000), with scores ranging from 1 (lowest) to 10 (highest).

2.4. Data analyses

2.4.1. Random intercept cross-lagged panel models

We ran five RI-CLPMs (Hamaker et al., 2015) to examine the following relationships over three timepoints: internalized weight bias with BMI (Model 1), internalized weight bias with psychological quality of life (Model 2), internalized weight bias with experiences of weight discrimination (Model 3), experiences of weight discrimination with BMI (Model 4), and experiences of weight discrimination with psychological quality of life (Model 5). Fig. 1 illustrates our RI-CLPM examining the relationships of internalized weight bias with BMI over time.

The RI-CLPM examines reciprocal within-person relationships among two or more variables over time by separating the total

Table 1
Participant characteristics.

Variable	
Race[a]	***n*(%)**
White	401 (62.36)
Black	62 (9.60)
Hispanic, Latino, or Spanish	51 (7.93)
Asian	23 (3.58)
More than one racial identity	97 (15.09)
Another racial identity	9 (1.40)
Gender identity [b]	
Cis-gender male	2806 (95.02)
Another gender identity	147 (4.98)
Sexual identity [b]	
Gay	2412 (81.68)
Bisexual	387 (13.11)
Another sexual identity	154 (5.22)
Sexual attraction [b]	
Exclusively men	1572 (52.23)
More often men than women	1161 (39.32)
Other	220 (7.45)
Sexual experiences [b]	
Exclusively men	1315 (44.53)
More often men than women	1394 (47.21)
Other	244 (7.45)
Country of Residence [b]	
United States of America	1294 (43.82)
United Kingdom	637 (21.57)
Australia	625 (21.16)
Canada	331 (11.21)
Age	***M*(*SD*)**
Time 1	37.70 (13.70)
Time 2	38.7 (13.60)
Time 3	40.40 (14.00)
Socioeconomic status	
Time 1	6.00 (1.90)
Time 2	5.99 (1.90)
Time 3	6.07 (1.90)
BMI	
Time 1	26.70 (6.20)
Time 2	27.00 (6.20)
Time 3	27.20 (6.30)
Experiences of weight discrimination	
Time 1	1.61 (0.69)
Time 2	1.59 (0.67)
Time 3	1.57 (0.67)
Internalized weight bias	
Time 1	3.39 (1.53)
Time 2	3.48 (1.54)
Time 3	3.44 (1.54)
Psychological quality of life	
Time 1	63.90 (16.70)
Time 2	64.10 (16.50)
Time 3	63.40 (16.60)

[a] We recorded race data for 643 participants due to a server-side error in data collection.
[b] N = 2953

variation in observed scores into between-person and within-person variation (Hamaker et al., 2015). Here, the random intercept controls for the between-person variation in an observed score (e.g., participants' average levels of internalized weight bias). The auto-regressive (e.g., BMI at Time 1 predicting BMI at Time 2) and cross-lagged (e.g., BMI at Time 1 predicting internalized weight bias at Time 2) effects represent within-person relationships among participants' deviations from their expected score for each construct at a given timepoint. Expected scores represent participants' own expected level of a construct at a given timepoint, accounting for the average deviation of their own trait level from the overall group mean (Hamaker et al., 2015). The results gleaned from RI-CLPMs can inform intervention development by elucidating what direction a given relationship operates at a within-person level, and by identifying which of the constructs under examination are the strongest predictors.

We built our models in R using the Lavaan package (Rosseel, 2012). In line with Hamaker and colleagues (2015), average scores represented internalized weight bias, experiences of weight discrimination, and BMI in our models. We standardized the observed scores for our psychological quality of life measure, as the total score values were too large to estimate our RI-CLPMs with, resulting in model convergence errors. Observed scores for each variable were loaded onto their own random intercept and within-person centred variables, with all factor loadings set to one. We constrained the strength of the cross-lagged and auto-regressive pathways to be equal across all timepoints; this is because, consistent with Hamaker and colleagues (2015), there is no theoretical reason to suspect the strength of these relationships would differ in strength over time, and the size of the intervals between our timepoints were approximately equal (median = 183 days for both gaps; Supplementary Fig. 2).

We included race, SES, age, and BMI (in models where BMI was not a dependent variable) as model covariates, as these constructs may affect the frequency with which people experience weight discrimination (Emmer et al., 2020; Major, Tomiyama, & Hunger, 2018), as well as their capacity to cope with these experiences (Emmer et al., 2020; Himmelstein, Puhl, & Quinn, 2017). The random intercepts for each model were regressed onto the between-person covariates (Mulder & Hamaker, 2020). We estimated our model using maximum likelihood estimation, and handled missing data with full information maximum likelihood (FIML) estimation.

We assessed model fit with chi squared tests, the root mean square error of approximation (RMSEA), comparative fit index (CFI), and standardised root mean square residual (SRMR). RMSEA values of 0.05, 0.08, 0.10, and >0.10 denote close, fair, mediocre, and unacceptable model fit (Browne & Cudeck, 1993). CFI values > 0.95 indicate good fit. SRMR values of less than 0.08 indicate good fit (Hu & Bentler, 1999). To determine whether relationships of weight stigma variables with outcomes are characterised by both within- and between-person variation, we also compared the fit of the traditional CLPM (which does not partition between- and within-person variation) with each of our RI-CLPMs using a chi square test.

2.4.2. Statistical power

We conducted a post-hoc Monte-Carlo simulation of the statistical power in our RI-CLPMs using MPlus version 8.1 and code adapted from Masselink et al. (2018) (also available on our Open Science Framework page). Our power analysis focused on our unadjusted model examining the cross-lagged relationships of BMI with internalized weight bias (Fig. 1). Within this model, our parameters of major interest were the regression coefficients representing the magnitude (strength) of our cross-lagged relationships. Results from 10,000 simulations revealed that our achieved power was 0.84 and 0.90 for regression coefficients of magnitude 0.09 and 0.25, respectively. Given the above, we believe we were sufficiently powered to detect small effects. Nonetheless, we acknowledge that estimating statistical power for RI-CLPMs is an ongoing area of research and note that our and others' assessment of our statistical power in this study may change in the future.

2.4.3. Missing data

The range of missing item-level data for each timepoint was: 3.1%–7.1% (M = 6.3%) at Time 1, 3.4%–6.0% (M = 5.1%) at Time 2, and 3.2%–5.4% (M = 4.98%) at Time 3. We conducted missing data analyses with the MissMech package in R (Jamshidian, Jalal, & Jansen, 2014). There was no evidence to reject the null hypothesis that data was missing at random at each timepoint.

Table 2
Pearson's correlations between continuous study variables across timepoints.

Variable		Age			BMI			SES			IWB			EWD			PQoL		
		T1	T2	T3	T1	T2	T3	T1	T2	T3	T1	T2	T3	T1	T2	T3	T1	T2	T3
Age	T1	-																	
	T2	.99**	-																
	T3	.99**	.99**	-															
BMI	T1	.14**	.13**	.13**	-														
	T2	.12**	.13**	.12**	.95**	-													
	T3	.12**	.11**	.13**	.92**	.96**	-												
SES	T1	.13**	.16**	.14**	-.14**	-.13**	-.16**	-											
	T2	.16**	.16**	.16**	-.10**	-.09**	-.13**	.74**	-										
	T3	.15**	.16**	.14**	-.12**	-.13**	-.13**	.72**	.74**	-									
IWB	T1	-.18**	-.19**	-.18**	.37**	.36**	.37**	-.26**	-.23**	-.25**	-								
	T2	-.18**	-.19**	-.19**	.35**	.37**	.38**	-.25**	-.25**	-.28**	.84**	-							
	T3	-.15**	-.17**	-.16**	.32**	.35**	.38**	-.27**	-.28**	-.28**	.81**	.85**	-						
EWD	T1	-.07*	-.08*	-.07*	.34**	.35**	.34**	-.29**	-.28**	-.27**	.51**	.47**	.47**	-					
	T2	-.09*	-.09**	-.08*	.35**	.33**	.38**	-.28**	-.28**	-.28**	.46**	.51**	.49**	.72**	-				
	T3	-.03	-.06*	-.06*	.35**	.34**	.34**	-.30**	-.28**	-.27**	.45**	.47**	.50**	.72**	.74**	-			
PQoL	T1	-.15**	-.16**	-.14**	.10**	.11**	.16**	-.43**	-.38**	-.41**	.54**	.54**	.52**	.44**	.41**	.42**	-		
	T2	-.18**	-.18**	-.17**	.09**	.11**	.14**	-.37**	-.40**	-.42**	.51**	.59**	.56**	.39**	.43**	.43**	.84**	-	
	T3	-.16**	-.15**	-.17**	.12**	.12**	.13**	-.41**	-.39**	-.43**	.52**	.56**	.58**	.39**	.40**	.45**	.81**	.86**	-

Notes: * = $p < .05$, ** = $p < .001$. BMI = body mass index. SES = socioeconomic status, IWB = internalized weight bias, EWD = experiences of weight discrimination, PQoL = psychological quality of life (high scores indicate lower psychological quality of life).

3. Results

3.1. Participant characteristics

Table 1 presents key participant characteristics at baseline and across timepoints. Differences in these characteristics between each timepoint are examined in our supplementary materials. Most participants identified as White, gay, and cis-gender. The average BMI across all timepoints was "overweight". Notably, average levels of experiences of weight discrimination were low across all timepoints.

3.2. Bivariate correlations among continuous variables

Table 2 presents correlations between all study variables.

3.3. RI-CLPMs examining reciprocal relationships among BMI, experiences of weight stigma, internalized weight bias, and psychological quality of life over time

Table 3 presents the results for our constrained RI-CLPMs (adjusted and unadjusted for covariates). There were no substantive differences in the results for our unadjusted and adjusted models. The full output for our constrained and unconstrained RI-CLPMs and CLPMs is available on our Open Science Framework page.

3.3.1. Internalized weight bias and BMI

Fig. 2 visualises the results for Model 1 and shows small cross-lagged effects of internalized weight bias on BMI, such that increases in participants' internalized weight bias (compared to their own average) were associated with increases in their BMI at the next timepoint. This relationship was reciprocal – increases in participants' BMI were associated with increases in their own average levels of internalized weight bias at the subsequent timepoint. The auto-regressive effects of both BMI and internalized weight bias were significant, such that increases in both of these constructs were associated with increases in the same construct at the next timepoint. The random intercepts were significantly correlated, indicating that participants with higher BMIs tended to report higher internalized weight bias. We note that, in our model including covariates, the unstandardized estimate for the pathway from BMI to internalized weight bias was significant, whereas the standardized was not (Fig. 2) – we discuss these discrepancies in our

Table 3
Estimates for our RI-CLPMs, unadjusted and adjusted for covariates (Race, SES, Age). Models 2, 3, and 5 also included BMI as a covariate.

		Model 2: I ↔ P [b]					Model 3: I ↔ E [c]				
		I → P	P → I	I → I	P → P	I ↔ P	I → E	E → I	I → I	E → E	I ↔ E
b	Unadjusted	0.08*	0.30**	0.20*	0.33**	0.74**	0.04	0.08	0.26*	0.03	0.49**
	Adjusted	0.08*	0.29**	0.19*	0.31**	0.52**	0.04	0.07	0.25*	0.03	0.30**
β^a	Unadjusted	0.12 (0.05)	0.19 (0.05)	0.19 (0.07)	0.33 (0.07)	0.49 (0.02)	0.11 (0.08)	0.05 (0.04)	0.24 (0.08)	0.09 (0.07)	0.47 (0.02)
		[0.01, 0.22]	[0.09, 0.29]	[0.05, 0.32]	[0.20, 0.46]	[0.44, 0.53]	[-0.04, 0.26]	[-0.03, 0.14]	[0.09, 0.39]	[-0.06, 0.24]	[0.42, 0.51]
	Adjusted	0.12 (0.05)	0.18 (0.05)	0.18 (0.07)	0.31 (0.07)	0.34 (0.02)	0.10 (0.07)	0.05 (0.04)	0.24 (0.09)	0.10 (0.07)	0.29 (0.02)
		[0.01, 0.22]	[0.09, 0.28]	[0.04, 0.31]	[0.18, 0.44]	[0.30, 0.38]	[-0.05, 0.22]	[-0.04, 0.13]	[0.09, 0.38]	[-0.05, 0.24]	[0.25, 0.33]

		Model 4: E ↔ B [d]					Model 5: E ↔ P [e]				
		E → B	B → E	E → E	B → B	E ↔ B	E → P	P → E	E → E	P → P	E ↔ P
b	Unadjusted	0.17	0.01	0.02	0.74**	1.33**	-0.02	0.05	0.02	0.35**	0.27**
	Adjusted	0.26	0.01	0.05	0.74**	1.21**	-0.01	0.06	0.02	0.33**	0.18**
β^a	Unadjusted	0.02 (0.02)	0.05 (0.12)	0.08 (0.07)	0.69 (0.05)	0.32 (0.02)	0.01 (0.04)	0.10 (0.07)	0.08 (0.07)	0.35 (0.07)	0.39 (0.02)
		[-0.03, 0.06]	[-0.19, 0.29]	[-0.06, 0.23]	[0.59, 0.80]	[0.27, 0.36]	[-0.08, 0.09]	[-0.05, 0.24]	[-0.06, 0.22]	[0.21, 0.49]	[0.35, 0.44]
	Adjusted	0.02 (0.02)	0.07 (0.11)	0.09 (0.07)	0.70 (0.05)	0.29 (0.02)	-0.01 (0.04)	0.11 (0.07)	0.08 (0.07)	0.33 (0.07)	0.26 (0.02)
		[-0.03, 0.06]	[-0.15, 0.29]	[-0.05, 0.23]	[0.60, 0.80]	[0.24, 0.33]	[-0.07, 0.09]	[-0.02, 0.24]	[-0.06, 0.21]	[0.18, 0.47]	[0.22, 0.30]

Notes: * $p < .05$, ** $p < .001$, [a] *β(SE)* [95% CI], [b] unadjusted: $\chi^2(5) = 2.31$, $p = .81$, adjusted: $\chi^2(37) = 36.33$, $p = .50$ [c] unadjusted: $\chi^2(5) = 3.56$, $p = .62$, adjusted: $\chi^2(37) = 33.89$, $p = .62$ [d] Fit: $\chi^2(5) = 4.37$, $p = .50$, adjusted: $\chi^2(33) = 36.10$, $p = .33$ [e] Unadjusted fit: $\chi^2(5) = 4.76$, $p = .45$, adjusted: $\chi^2(37) = 33.74$, $p = .62$ Extra fit indices for all models (adjusted and unadjusted) were: RMSEA =.00, CFI = 1.00, SRMR =.01. I = Internalized weight bias, P = psychological quality of life, E = experiences of weight discrimination, B = BMI, ↔ = random intercept. Higher psychological quality of life scores denote poorer psychological quality of life.

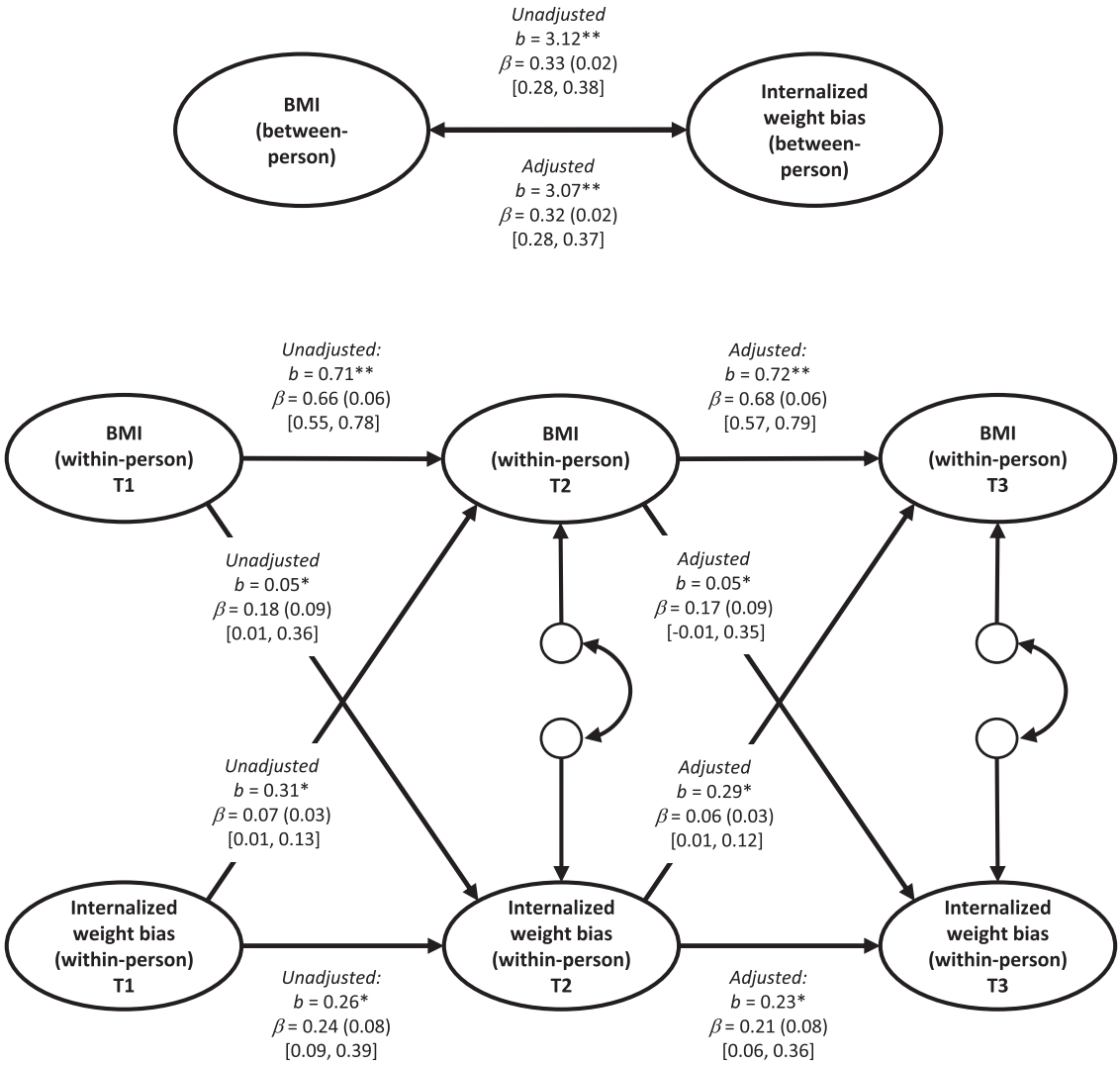

Fig. 2. Results for our constrained RI-CLPMs modelling the relationships of BMI and internalized weight bias over three timepoints. Unadjusted estimates are presented on the left, and estimates adjusted for SES, race, and age on the right. Model fit statistics: Unadjusted: $\chi^2(5) = 3.61$, $p = .61$, Adjusted: $\chi^2(5) = 32.48$, $p = .49$. Alternative fit statistics for unadjusted and adjusted models were: RMSEA =.00, CFI = 1.00, SRMR =.01. Standard errors are in round brackets, and 95% confidence intervals around the standardized regression coefficients are in square brackets. * $p < .05$, ** $p < .001$. We note that the adjusted standardized and unstandardized estimates for the cross-lagged effect of internalized weight bias on BMI differed in significance – we discuss these differences in our Supplementary Materials (online).

supplementary materials. Our constrained models with and without covariates demonstrated excellent fit. There was no significant difference in the fit between our constrained and unconstrained RI-CLPM (p =.74). The RI-CLPM fit significantly better than the CLPM ($p < .001$).

3.3.2. Internalized weight bias and psychological quality of life

As shown in Table 3, there were small significant cross-lagged effects of psychological quality of life on internalized weight bias: within-person increases in internalized weight bias were associated with small decreases in psychological quality of life at the previous timepoint. Within-person reductions in psychological quality of life were also associated with increased levels of internalized weight bias at the preceding timepoint. The auto-regressive effects of both psychological quality of life and internalized weight bias were significant. The random intercepts were strongly correlated, indicating that participants with higher levels of internalized weight bias tended to report lower psychological quality of life. The model fit for our constrained RI-CLPMs examining relationships among internalized weight bias and psychological quality of life, both adjusted and unadjusted for covariates, was excellent. There was no significant difference in the fit between our constrained and unconstrained RI-CLPMs (p =.71). The RI-CLPM fit significantly better than the CLPM ($p < .001$).

3.3.3. Internalized weight bias and experiences of weight discrimination

As shown in Table 3, there were no significant cross-lagged effects in either direction. The autoregressive effects for internalized weight bias were significant, such that increases in participants' own average levels of internalized weight bias at one time were associated with small increases in their level of internalized weight bias at the next timepoint. The random intercepts for Model 2 were significantly correlated, indicating that participants with higher trait internalized weight bias tended to report more frequent experiences of weight discrimination. The autoregressive effects for experiences of weight discrimination were not significant. The model fit for our constrained models assessing the relationship of internalized weight bias with experiences of weight discrimination was excellent. There was no significant difference in the fit between our constrained and

unconstrained RI-CLPM (p =.95). The RI-CLPM fit significantly better than the CLPM (p < .001).

3.3.4. Experiences of weight discrimination and BMI

As shown in Table 3, there were no significant cross-lagged effects in either direction. The autoregressive effects for BMI were significant, such that increases in participants' own average BMI were associated with higher BMIs at the subsequent timepoint. The autoregressive effects for experiences of weight discrimination were not significant. The random intercepts in Model 4 were significantly correlated, indicating that participants with higher trait internalized weight bias tended to report with more frequent experiences of weight discrimination. The model fit for our constrained models adjusted and unadjusted for covariates was excellent. The unconstrained RI-CLPM fit slightly better than the constrained RI-CLPM (p = .004). The RI-CLPM fit significantly better than the CLPM (p < .001).

3.3.5. Experiences of weight discrimination and psychological quality of life

As shown in Table 3, there were no significant cross-lagged effects in either direction. The autoregressive effect for experiences of weight discrimination was not significant. Experiences of weight discrimination and psychological quality of life were significantly associated between-persons. The autoregressive effect of psychological quality of life on itself was significant. The model fit for our constrained models of experiences of discrimination and psychological quality of life, adjusted and unadjusted for covariates, was excellent. There was no significant difference in the fit between our constrained and unconstrained RI-CLPM (p =.71). The RI-CLPM fit significantly better than the CLPM (p < .001).

4. Discussion

In a large cohort of sexual minority men, we examined longitudinal within-person relationships of internalized weight bias, experiences of weight discrimination, BMI, and psychological quality of life at three time points over 12 months. Our within-person hypotheses for internalized weight bias were supported, finding significant cross-lagged relationships of internalized weight bias on BMI and psychological quality of life. Further, these relationships were bidirectional, such that within-person decreases in psychological quality of life, and increases in BMI, were associated with increased internalized weight bias at the next timepoint. We did not find significant within-person relationships of experiences of weight discrimination with internalized weight bias, BMI or psychological quality of life. Our between-person hypotheses were supported, finding that higher average experiences of weight discrimination and internalized weight bias were both associated with higher average BMIs and lower psychological quality of life scores. Experiences of weight discrimination and internalized weight bias were positively associated between-persons. Our results were consistent when controlling for SES, BMI, age, and race.

4.1. Between-person associations of internalized weight bias and experiences of discrimination with psychological quality of life and BMI

Between-person associations were significant across all five of our models. Internalized weight bias was positively associated with psychological quality of life and BMI, indicating that those with higher BMIs also tend to have higher average internalized weight bias. We also found positive between-person associations of internalized weight bias and psychological quality of life: higher average internalized weight bias was associated with lower average psychological quality of life. Experiences of weight discrimination were associated with internalized weight bias, BMI, and psychological quality of life between-persons only, indicating that those who experience more weight discrimination than the average also tended to report higher internalized weight bias, higher BMIs, and poorer psychological quality of life. Together, these results corroborate existing studies that evidence between-person associations of experiences of weight discrimination and internalized weight bias with psychological quality of life and BMI across samples with diverse body weights (for reviews, see Pearl & Puhl, 2018; Emmer et al., 2020).

4.2. Feedback loops of internalized weight bias with psychological quality of life and BMI

Model 1 revealed reciprocal feedback loops of internalized weight bias with BMI. Within-person increases in internalized weight bias were associated with small BMI increases at the next timepoint, which were associated with subsequent small increases in internalized weight bias. Thus, weight gain may make people more prone to internalizing negative weight-based stereotypes and self-devaluation, facilitating further weight gain. This accords with findings from Pearl and colleagues (2021), wherein internalized weight bias was positively associated with 'eating to cope' and 'gym avoidance' among adults enrolled in weight management. Relatedly, Major and colleagues (2018) note that being more aware of negative weight stereotypes following one's own weight gain may make individuals avoid 'health-related' behaviours out of fear of facing stigmatization (i.e., *stigma avoidance*), thereby facilitating further weight gain (see Section 4.4. for a detailed discussion of mechanisms).

In Model 2, we found that increases in internalized weight bias were associated with small reductions in psychological quality of life at the subsequent timepoint. This relationship was reciprocal, with decreases in psychological quality of life associated with small increases in internalized weight bias at subsequent timepoints. Thus, internalized weight bias may relate to poorer psychological wellbeing in a reciprocal manner, building upon previous work that has examined internalized weight bias as a mediator or moderator of the relationship between experiences of weight discrimination and poorer mental health (Curll & Brown, 2020; Hayward et al., 2018). Our results support previous suggestions that these relationships may also operate in the opposite direction to what is typically hypothesized – that is, experiencing reduced psychological wellbeing may also make people more prone to internalizing weight bias over time (Hayward et al., 2018; Pearl & Puhl, 2018; Pearl, Puhl, Lessard, Himmelstein & Foster, 2021).

4.3. Experiences of weight discrimination are not longitudinally predictive of outcomes

We did not find evidence of significant within-person relationships of experiences of weight discrimination with any outcome. The mean levels of experiences of weight discrimination were low in our study (falling between "Never/Not at all" and "Rarely"; Table 1), perhaps because the average BMI of our sample was within the "overweight" BMI category. Among men, "overweight" BMIs may be attributable to greater levels of muscle mass, rather than body fat, which may engender less weight-related discrimination. Our results may have differed men in the "obese" and "underweight" BMI categories, who evidently face the highest burden of weight stigma experiences among male samples (Griffiths et al., 2018a). We note that the relatively low variability we observed in our participants' weight discrimination scores may have weakened our statistical

power to detect very small- or small-sized relationships; though, as previously described in Section 2.4.2, we were reasonably confident in the amount of statistical power available to us. Alternatively, experiences of weight discrimination may manifest in outcomes within a shorter timeframe than we assessed; assessing these relationships on a daily basis via ecological momentary assessment methodology may reveal stronger within-person associations of weight discrimination with outcomes (c.f. Vartanian et al., 2018).

We also note that our measure of weight discrimination does not capture the more passive, covert forms of weight discrimination that people may experience. Consider, for example, the everyday microaggressions that fat people face, from clothing brands and furniture that do not accommodate their size, to medical professionals prescribing weight loss or 'lifestyle changes' as a default solution to their aliments without exploring alternative causes (for a useful review, see Phelan, Burgess, Yeazel, Hellerstedt, Griffin & van Ryn, 2015). In contrast, the EDS explicitly asks for people to recall experiences of more overt discrimination (e.g., "You are threatened or harassed"; Williams et al., 1997). Our results may have differed had we used a measure that encompassed both the more covert, passive forms of weight discrimination, as well as the more active or overt forms.

Speaking specifically to the non-significant within-person relationships of internalized weight bias with experiences of discrimination, these results may indicate that experiencing overt discrimination is not a prerequisite to internalizing weight bias. It is likely that weight bias is not internalized exclusively as a product of overt discrimination, but also from the passive everyday experiences of living in a world where fatness is devalued. As one telling example, consider the context of the COVID-19 pandemic, where lockdowns implemented worldwide to curb the spread of the virus have incited significant concerns of weight gain in public (and private) discourse – see, for example, media references to "hid[ing] those lockdown pounds" (Murphy, 2021) or advice about how to lose the "quarantine-15" (a play on the phrase "freshman-15"; Gordon, 2021). These messages, although not explicitly discriminatory, nonetheless perpetuate weight stigma by implying weight gain and fatness are something to be "fixed". For a useful theoretical model of how cultural pressures may facilitate internalized weight bias and contribute to negative health outcomes within the context of COVID-19, see Pearl and Schulte (2021).

Relatedly, we note that part of this study took place throughout 2020; as such, the rhetoric surrounding COVID-19 weight gain may have facilitated higher levels of internalized weight bias in our sample at times 2 and 3. Notably, the pandemic provides an almost ideal context for examining changes in weight stigma and its interrelations with different constructs over time. In cross-lagged panel models, the causal mechanisms (sometimes referred to as 'impulses'; see Zyphur et al., 2020 for a useful explanation) are assumed to be some sort of external event (e.g., the pandemic) that facilitates a momentary change in an independent variable (e.g., someone's level of internalized weight bias). An RI-CLPM examines the extent to which the momentary change in that variable facilitates a change in another variable (e.g., someone's psychological wellbeing) at a subsequent timepoint. In our study, the pandemic and its associated weight gain rhetoric are plausible candidate impulses, which may have driven momentary changes in our study variables and created an ideal context for examining changes in these phenomena over time.

Taken together, our within-person results suggest that internalized weight bias may be the most potent mechanism in the relationship of weight stigma with its associated outcomes, including BMI and psychological quality of life. These findings have notable implications for intervention development, which we detail in Section 4.5.

4.4. Modelling complex longitudinal relationships of weight stigma and outcomes

We note that our models only represent relationships among pairs of phenomena. The relationships revealed in our study may operate in tandem at a within- and between-person level, and in combination with various other mechanisms – for example, experiences of weight discrimination may predict poorer psychological quality of life through internalized weight bias (Austen et al., 2020), concerns about experiencing weight stigma (Hunger & Major, 2015), or the coping responses employed after experiencing weight discrimination (Hayward et al., 2018). As one example of how within- and between-person processes may operate together to manifest outcomes over time, experiences of weight discrimination may predict psychological distress at a within-person level, but only in those with high average levels of internalized weight bias (i.e., a between-person moderator; c.f. Vartanian et al., 2018). Complex longitudinal models of within- and between-person relationships of weight stigma with outcomes are required to identify populations who are most impacted by these phenomena and to highlight precise intervention targets.

4.5. Theoretical and practical implications

Our study provides important insights for intervention development and implementation. Our within-persons results can be harnessed to understand how weight stigma may operate at an individual level – specifically, internalizing weight bias can lead to weight gain and/or poorer psychological wellbeing over time, and vice-versa. This supports emerging interventions that prioritise internalized weight bias as an intervention target (Griffiths et al., 2018b; Pearl et al., 2020), and highlights psychological wellbeing as another worthy target. Interventions may benefit by providing strategies to bolster overall psychological wellbeing, as well as targeting internalized weight bias directly.

The bidirectional relationships of BMI with internalized weight bias provide impetus to determine the precise mechanisms through which these phenomena are associated – for example, whether internalized weight bias is a barrier to engagement in the health-related behaviours (e.g., exercise; Pearl et al., 2021) that may act as a buffer to the negative health-related outcomes of weight stigma. Importantly, we do not propose that BMI is an intervention target, given the risks associated with weight loss-related phenomena (e.g., maladaptive weight suppression and eating disorder development; Stice, Rohde, Shaw & Desjardins, 2020).

Our study also adds to growing literature (e.g., Austen et al., 2020; Griffiths et al., 2018; Puhl et al., 2019) that indicates sexual minority men are subject to, and face adverse consequences from, weight stigma. This emphasizes the importance of continuing to provide dedicated investigations of weight stigma and its outcomes across demographic groups underrepresented in weight stigma research – for example, racial, sexual, and gender minority identities. For interventions to be effective across a range of demographic groups, studies must continue to broaden their focus beyond those populations that form the focus of the majority of existing research (i.e., individuals who are White, heterosexual, female, and of high SES; Pearl & Puhl, 2018; Emmer et al., 2020).

Lastly, our constrained RI-CLPMs were an excellent fit for our data, and, in most cases, fit better than the traditional CLPM, indicating that the relationships of weight stigma with BMI and psychological quality of life are characterised by distinct within- and between-person processes. Future research should continue to separate these sources of variation, given that causal relationships of weight stigma with its outcomes within-persons may be confounded by between-person variation (Hamaker et al., 2015).

4.6. Limitations and future directions

The key limitation of this study is the limited generalisability of our results: our sample comprised mostly White, cis-gender men with an average "overweight" BMI across all timepoints – thus, our results may not generalise to populations that do not share these demographic characteristics. Relatedly, our sample was recruited from Grindr – a platform noted for its particularly heavy emphasis on appearance and physique (Anderson et al., 2018). The appearance pressures evident on Grindr may have manifested in heightened levels of internalized weight bias or experiences of weight discrimination compared to sexual minority men who do not use this platform. We also note that we did not explore differences in results across sexual or gender identities. The nature of appearance pressures likely varies across specific identities under the 'sexual minority men' label – for example, cis-gender, transgender, and non-binary men will inevitably face varying and unique appearance concerns, as well as intersectional stressors that contribute to mental health disparities (Pröderl & Tremblay, 2015). Future research should aim to delineate experiences of weight stigma among specific identities under the 'sexual minority men' umbrella, as well as other sexual and gender minority identities. Finally, our height and weight data would likely be more accurate had objective measures been used; however, we note that self-report measures of weight and height are nonetheless strongly correlated (see Methods).

4.7. Conclusion

Our results suggest that increases in internalized weight bias are associated with future weight gain and diminished psychological wellbeing among sexual minority men. The presence of reciprocal feedback loops of internalized weight bias with reduced psychological wellbeing and weight gain emphasizes the need for interventions that target both internalized weight bias and psychological quality of life. Internalized weight bias appears to be a potent and promising intervention target.

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CRediT authorship contribution statement

Emma Austen: Conceptualization, Formal analysis, Data curation, Visualisation, Writing – original draft, Writing – review & editing. **Scott Griffiths:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.bodyim.2021.10.006.

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